

AI 683 – Statistical Learning

Kaggle Competition Project: Playground Series – Season 5, Episode 11 Predicting Loan Payback

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Problem and Data

The task is a **binary classification problem**: predict whether a loan will be paid back.

- **Dataset**: Tabular data with applicant income, co-applicant income, loan amount, credit score, and categorical attributes.
- test.csv, train.csv, submission.csv dataset provided.
- **Target variable**: Loan status (paid back vs. default).
- **Evaluation metric**: ROC–AUC, as specified by the competition.

Methods

- **Preprocessing**:
 - Numeric features → median imputation + scaling.
 - Categorical features → most-frequent imputation + one-hot encoding.
- **Feature engineering**:
 - total_income (applicant + co-applicant).
 - loan_income_ratio (loan amount / total income).
 - Log transforms for skewed variables.
 - Credit score buckets.
- **Models**:
 - Baseline: Logistic Regression.
 - Tree-based: Random Forest, LightGBM, XGBoost, CatBoost.
 - Ensemble: Logistic Regression meta-model stacking LGBM, XGB, and CatBoost.
- **Validation**: Stratified 5-fold cross-validation for robust OOF estimates.

- **Tuning:** Grid search for Logistic Regression; tuned boosting parameters for LGBM/XGB/CatBoost.

Results

- **Base model OOF AUCs:**
 - LightGBM: 0.92197
 - XGBoost: 0.92145
 - CatBoost: 0.92147

OOF AUC: 0.9223

Leaderboard

- Previous best: 0.92008
- Current best: 0.92181

Conclusion

Using systematic preprocessing, feature engineering, and model comparison, we obtained high and stable scores on the loan payback prediction exercise. As the individual gradient boosting models had reached an OOF ROC -AUC of approximately 0.9215 a stack of CatBoost, LightGBM and XGBoost with a Logistic Regression meta-model boosted the ensemble to 0.9223 OOF AUC.

The screenshot shows the Kaggle interface for the 'Predicting Loan Payback' competition. The 'Leaderboard' tab is selected, displaying a submission by 'Bhakti Vyas123' with a score of 0.92181. The submission is labeled 'submission.csv' and was submitted 2 days ago. A search bar at the top of the leaderboard allows filtering by team name, currently showing 'bhakti vya'. The leaderboard table lists the top entry with rank 1873, team 'Bhakti Vyas123', score 0.92181, 2 entries, and a join link. A notification at the bottom states 'Your Best Entry! Your most recent submission scored 0.92181, which is an improvement over your previous score of 0.92008. Great job!' with a 'Tweet this' button.

Predicting Loan Payback Submit Prediction

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✓ **submission.csv** Score: 0.92181
Submitted by Bhakti Vyas123 · Submitted 2 days ago
Private score:

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#	Team	Members	Score	Entries	Last	Join
1873	Bhakti Vyas123		0.92181	2	2d	

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#	Team	Members	Score
1873	Bhakti Vyas123		0.92181

Your Best Entry!

Bhakti Vyas123

×

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8d